



Lecture 32

Residuals

Announcements

- **Homework 10** due Wednesday, 4/10
 - **Lab 9 & Project 2** due Friday, 4/12
 - **OH wait times** will likely be longer than usual due to the project deadline. If you have a question that you think other students will have as well, please make a public Ed post! Just make sure you are not sharing any of your code.
 - **Professor Sahai** will also **not be holding office hours** today. Instead, he will be online to answer questions via Ed or email during his regularly scheduled OH time (3-5pm).
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Weekly Goals

- **Today**

- Least squares: finding the "best" line for a dataset
- Residuals: analyzing mistakes and errors

- Thursday

- Regression inference
- Uncertainty in the slope & intercept

- Friday

- No lecture!
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Least Squares

(Demo)

Error in Estimation

- **error = actual value – estimate**
 - Typically, some errors are positive and some negative
 - To measure the rough size of the errors
 - **square** the **errors** to eliminate cancellation
 - take the **mean** of the squared errors
 - take the square **root** to fix the units
 - **root mean square error** (rmse)
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Numerical Optimization

- Numerical minimization is approximate but effective
 - Lots of machine learning uses numerical minimization
 - If the function `mse(a, b)` returns the mse of estimation using the line “estimate = $ax + b$ ”,
 - then `minimize(mse)` returns array `[a0, b0]`
 - `a0` is the slope and `b0` the intercept of the line that *minimizes* the mse among lines with arbitrary slope `a` and arbitrary intercept `b` (that is, among all lines)
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Least Squares Line

- Minimizes the root mean squared error (rmse) among all lines
 - Equivalently, minimizes the mean squared error (mse) among all lines
 - Names:
 - “Best fit” line
 - Least squares line
 - Regression line
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Errors and Residuals

Residuals

- Error in regression estimate
- One residual corresponding to each point (x, y)
- **residual**
 - = observed y - regression estimate of y**
 - = observed y - height of regression line at x
 - = vertical distance between the point and the best line

(Demo)

Regression Diagnostics

Example: Dugongs



(Demo)

Residual Plot

A scatter diagram of residuals

- Should look like an unassociated blob for linear relations
 - But will show patterns for non-linear relations
 - Used to check whether linear regression is appropriate
 - Look for curves, trends, changes in spread, outliers, or any other patterns
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Properties of residuals

- Residuals from a linear regression **always** have
 - **Zero** mean
 - (so **rmse = SD of residuals**)
 - **Zero** correlation with x
 - **Zero** correlation with the fitted values
 - These are all true **no matter what the data look like**
 - Just like deviations from mean are zero on average
(Demo)
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Discussion Questions

How would we adjust our regression line...

- if the average residual were 10?
 - if the residuals were positively correlated with x ?
 - if the residuals were above 0 in the middle and below 0 on the left and right?
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